UNIT 3 · 7 hours

Bode Plot: Construction



Frequency Domain Analysis

Control Systems (22EEG53)
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NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

- Explain the Bode plot and decibel scale
- Plot the asymptotic magnitude and phase of standard factors
- Construct the Bode plot of a given G(s)
- Determine the slope and corner frequencies

Syllabus context

Frequency domain analysis: frequency domain specifications peak resonance, resonant frequency, bandwidth and cut-off rate; correlation between time and frequency responses for second-order systems; Bode plot, gain margin and phase margin.

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What is a Bode plot?

Two plots: magnitude $20 \log|G(j\omega)|$ in dB and phase $\angle G(j\omega)$ in degrees, both against $\log \omega$.

Log scale turns products into sums: each factor of G(s) adds straight-line segments.

Octave and decade: frequency ratio 2 and 10; −20 dB/dec = −6 dB/octave.

Uses: stability margins, bandwidth, compensator design.

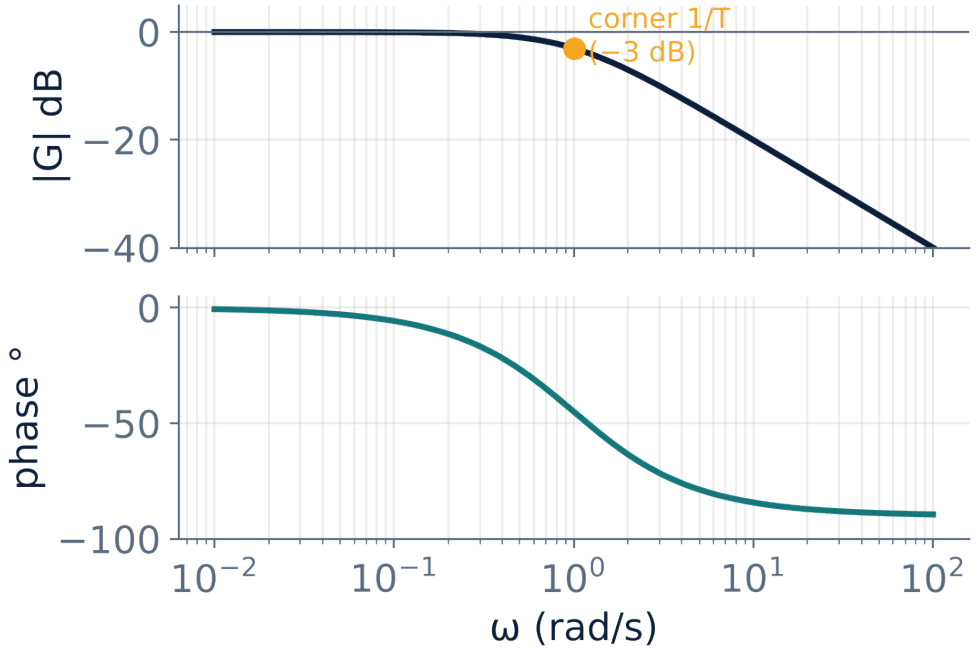
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Standard factors

Factor	Magnitude	Phase
Constant K	20 log K (flat)	0° (K > 0)
Integrator 1/(jω)ⁿ	−20n dB/dec through 0 dB at ω = 1	−90n°
Differentiator (jω)	+20 dB/dec	+90°
Simple lag 1/(1 + jωT)	0 dB then −20 dB/dec after corner 1/T	0° → −90° (−45° at 1/T)
Simple lead (1 + jωT)	0 dB then +20 dB/dec after 1/T	0° → +90°
Quadratic lag	−40 dB/dec after ωn; peak near ωn	0° → −180°

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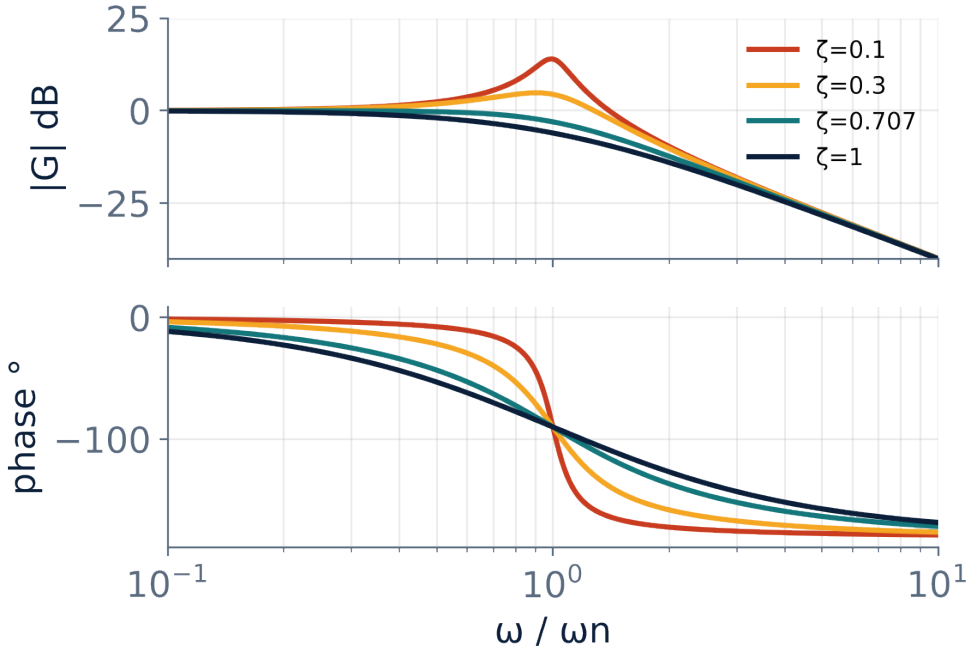
First-order lag: 1/(1 + jωT)



- Corner frequency ωc = 1/T.
- Below ωc: 0 dB; above: −20 dB/dec.
- At ωc exact magnitude = −3 dB, phase = −45°.
- Phase goes 0° → −90°.

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Second-order lag with damping



- Corner at ωn: slope changes by −40 dB/dec.
- Peak near ωn for ζ < 0.707: Mr dB.
- Phase from 0° to −180°, passing −90° at ωn.
- Smaller ζ → sharper resonance and phase change.

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Construction procedure

1

Standard form

Write G(jω) in time-constant form with K as gain.

2

Corner frequencies

List all 1/T values; arrange in increasing order.

3

Initial slope

−20N dB/dec at low ω (N = type), through 20 log K at ω = 1.

4

Change slope

At each corner: poles −20 dB/dec, zeros +20 dB/dec (−40 for quadratic poles).

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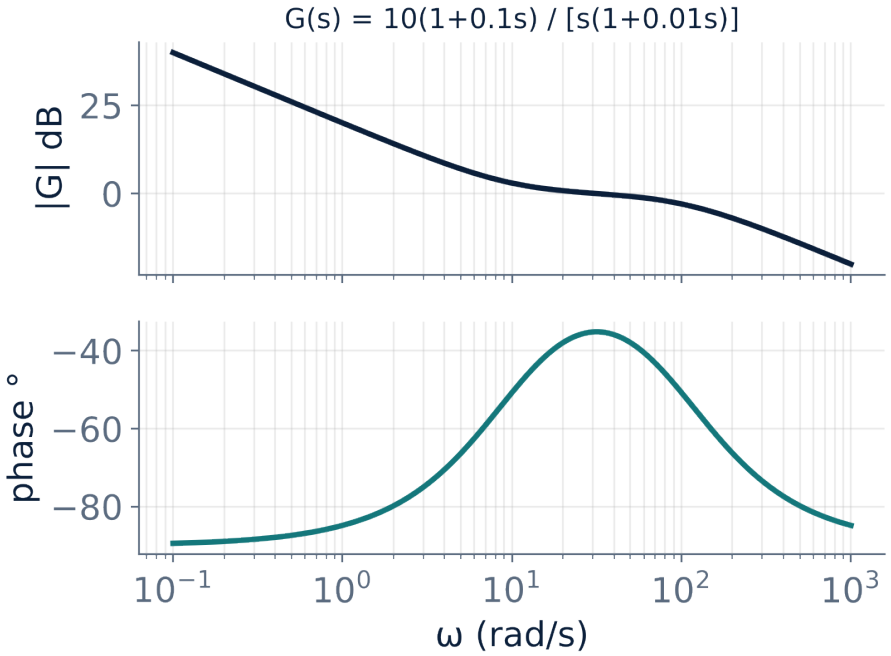
Phase

Add the phase of each factor point by point.

Initial magnitude at ω = 1 rad/s is 20 log K for a type 1 system.

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Example: G = 10(1 + 0.1s) / [s(1 + 0.01s)]



- Type 1, K = 10: starts at −20 dB/dec; 0 dB at ω = 10 (20 log 10 = 20 dB at ω = 1).
- Zero at ω = 10: slope → 0 dB/dec.
- Pole at ω = 100: slope → −20 dB/dec.
- Phase: −90° → ≈ −35° → −90°.

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Numerical: asymptotic magnitude

Problem G(s) = 100/[s(1 + 0.1s)]. Find the corner frequency, slopes and the gain crossover frequency.

- Type 1, K = 100: at ω = 1, 20 log 100 = 40 dB with slope −20 dB/dec
- Corner at ω = 1/0.1 = 10 rad/s: magnitude there = 40 − 20 log 10 = 20 dB; slope → −40 dB/dec
- 0 dB reached: 20 dB at 10 needs 20/40 = 0.5 decade more → ω ≈ 10 × 10^0.5

Answer: Gain crossover ≈ 31.6 rad/s (exact ≈ 31.5)

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Key takeaways

- Bode: 20 log|G| (dB) and phase vs log ω
- Factor slopes: ±20 dB/dec per pole/zero, integrator −20
- Corner frequency = 1/T; at corner −3 dB, −45°
- Initial slope = −20 × type

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Practice questions

Q1. Draw the asymptotic Bode magnitude plot of $G = 20/[(1 + 0.5s)(1 + 0.05s)]$.

Q2. State the slope contributions of poles and zeros.

Q3. Find the phase at $\omega = 1/T$ for $1/(1 + sT)$.

Q4. Construct the Bode plot for $G = 10/[s(s + 10)]$.